README

> 本文主要讨论利用UMAT子程序进行IV型复合储氢瓶爆破压强下的渐进失效分析，分别给出三个子程序并给出对应模型案例
>
> - 只用来判断hashin准则的子程序 **onlyhashin.for**
> - 刚度直接折减的hashin子程序 **hashin_constant.for**
> - 基于CDM的渐进失效hashin子程序 **hashin.for**

要注意的问题

> UMAT子程序利用雅可比矩阵进行迭代本身难以收敛，尤其是在处理爆压失效方面

- IV型瓶使用的塑料内胆（HDPE或PA6）弹性模量通常过小，最大的PA6一般为1800MPa左右，这使得气瓶在压力不断增大中，往往是内胆区域的单元变形过大，尤其是封头区，而导致不收敛并非是由于纤维失效过多

  > 目前的建议是，要么转为vumat做这种爆破失效，要么**适当提高内胆弹性模量**

- 本次子程序都是针对复合缠绕层的，缠绕层所用单元都为C3D8，如果使用C3D8R，建议把沙漏控制打开，避免报错，**建议使用C3D8**
- 关于网格划分，本次的气瓶缠绕层都是用WCM进行铺层且自动划分，尽量的使其规整，不要出现那种切碎或者尖锐的网格；关于封头与内胆，建议是**以一个整体分不同区域赋予材质实现网格共节点**，当然本文有些模型把封头与内胆作为不同部件划分，使用tie连接
- 本文模型及所用材料属性皆来自于不同网络文献，结果不一定准确，另外所用hashin公式也来自于其他文献，对于不同公式及不同材料属性，其对应的收敛性都难于保证
- 本文子程序同样适用于三型瓶，且其由于是金属内胆收敛性往往更好，对于IV型瓶，如果想要好的结果，建议只拿子程序做工作压强（比如常见20MPa或者70Mpa）,合格爆炸压强通常为工作压强的2.25倍

3D Hashin 公式及渐进损伤理论

> 本次子程序所用公式使用**应力判断**，事实上，可能使用应变更合适（可以参考其他使用应变的hashin准则子程序）

3d hashin准则

所用准则来自文献“Modelling of the low-impulse blast behaviour of fibre–metal laminates based on different aluminium alloys”

> 注意原文是用vumat做复材抗爆分析，本umat子程序借用了其判决式和退化准则

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Modelling of the low-impulse blast behaviour of fibre–metal laminates based on different aluminium alloys

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Fiber Failure in Tension

When $\sigma_{11} > 0$

$$F_f = \left(\frac{\sigma_{11}}{X_T}\right)^2 + \frac{1}{S_{12}^2}(\tau_{12}^2 + \tau_{13}^2)$$

Fiber Failure in Compression

When $\sigma_{11} < 0$

$F_f = \left(\frac{-\sigma_{11}}{X_C}\right)^2$

Matrix Failure in Tension

When $\sigma_{22} + \sigma_{33} > 0$

$F_m = \frac{1}{Y_T^2}(\sigma_{22} + \sigma_{33})^2 + \frac{1}{S_{23}^2}(\tau_{23}^2 - \sigma_{22}\sigma_{33}) + \frac{1}{S_{12}^2}(\tau_{12}^2 + \tau_{13}^2)$

Matrix Failure in Compression

When $\sigma_{22} + \sigma_{33} < 0$

$F_m = \frac{1}{Y_C}\left[\left(\frac{Y_C}{2S_{23}}\right)^2 - 1\right](\sigma_{22} + \sigma_{33}) + \frac{1}{4S_{23}^2}(\sigma_{22} + \sigma_{33})^2 + \frac{1}{S_{23}^2}(\tau_{23}^2 - \sigma_{22}\sigma_{33}) + \frac{1}{S_{12}^2}(\tau_{12}^2 + \tau_{13}^2)$

损伤演化矩阵

当损伤发生后，材料刚度开始退化，应力应变关系由 定义：

$\sigma = C(d) \cdot \epsilon$

其中，$C(d)$ 是一个 6×6 的对称受损矩阵，其非零项完整展开如下：

- $C_{11} = (1 - d_f)E_1(1 - v_{23}v_{32})\Gamma$
- $C_{22} = (1 - d_f)(1 - d_m)E_2(1 - v_{13}v_{31})\Gamma$
- $C_{33} = (1 - d_f)(1 - d_m)E_3(1 - v_{12}v_{21})\Gamma$
- $C_{12} = (1 - d_f)(1 - d_m)E_1(v_{21} + v_{31}v_{23})\Gamma$
- $C_{23} = (1 - d_f)(1 - d_m)E_2(v_{32} + v_{12}v_{31})\Gamma$
- $C_{13} = (1 - d_f)(1 - d_m)E_1(v_{31} + v_{21}v_{32})\Gamma$
- $C_{44} = (1 - d_f)(1 - s_{mt}d_{mt})(1 - s_{mc}d_{mc})G_{12}$
- $C_{55} = (1 - d_f)(1 - s_{mt}d_{mt})(1 - s_{mc}d_{mc})G_{23}$
- $C_{66} = (1 - d_f)(1 - s_{mt}d_{mt})(1 - s_{mc}d_{mc})G_{13}$

此处，全局纤维损伤变量 d_f、基体损伤变量 d_m 以及常数 Γ 的定义为：

- $d_f = 1 - (1 - d_{ft})(1 - d_{fc})$
- $d_m = 1 - (1 - d_{mt})(1 - d_{mc})$
- $\Gamma = 1/(1 - v_{12}v_{21} - v_{23}v_{32} - v_{13}v_{31} - 2v_{21}v_{32}v_{13})$

当损伤发生后，材料刚度开始退化，应力应变关系由 定义：

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- $C_{12} = (1 - d_f)(1 - d_m)E_1(v_{21} + v_{31}v_{23})\Gamma$
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- $C_{55} = (1 - d_f)(1 - s_{mt}d_{mt})(1 - s_{mc}d_{mc})G_{23}$

- $C_{66} = (1-d_f)(1-s_{mt}d_{mt})(1-s_{mc}d_{mc})G_{13}$

此处，全局纤维损伤变量 d_f、基体损伤变量 d_m 以及常数 Γ 的定义为（公式 10）：

- $d_f = 1-(1-d_{ft})(1-d_{fc})$
- $d_m = 1-(1-d_{mt})(1-d_{mc})$
- $\Gamma = 1/(1-v_{12}v_{21}-v_{23}v_{32}-v_{13}v_{31}-2v_{21}v_{32}v_{13})$

  注：文献中建议的剪切模量折减系数为 $s_{mt}=0.9$ 和 $s_{mc}=0.5$

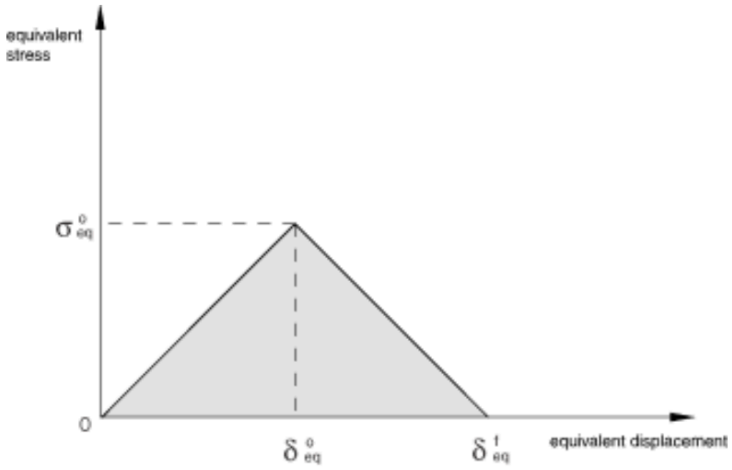
> 注意，Thuc的退化准则中剪切相的折减考虑了纤维的影响即乘了（1-df）,实际运行中发现这会大幅降低气瓶的最终爆压即不容易收敛，毕竟复合缠绕层纤维是主要受力对象而且用的C3D8可能会有“剪切自锁”，所以实际hashin.for子程序对退化矩阵进行了修改，剪切相去掉（1-df）,一般的hashin退化矩阵剪切都没直接考虑纤维影响，但hashin_constant的公式还是完全照文献

如果是只使用hashin_constant.for子程序，到这里就结束了

基于断裂能（能量）控制的线性软化规律

abaqus 2024版本的帮助手册有专门介绍的（主要针对二维hashin）

Figure 1. Equivalent stress versus equivalent displacement.



其中（接上部分）：

$$d = \frac{\delta_{eq}^{f}\left(\delta_{eq}-\delta_{eq}^{0}\right)}{\delta_{eq}\left(\delta_{eq}^{f}-\delta_{eq}^{0}\right)}$$

这里采用的等效位移及等效应力的计算参考文献“An Enhanced Progressive Damage Model for Laminated Fiber-Reinforced Composites Using the 3D Hashin Failure Criterion: A Multi-Level Analysis and Validation”

materials MDPI

Article

An Enhanced Progressive Damage Model for Laminated Fiber-Reinforced Composites Using the 3D Hashin Failure Criterion: A Multi-Level Analysis and Validation

Yichen Zhang [1], Wim Van Paepegem [2] and Wouter De Corte [1,*]

Failure Mode	Equivalent Displacement (δeqi)	Equivalent Stress (σeqi)
FT	$L_c\sqrt{\langle\varepsilon_{11}\rangle^2+\varepsilon_{12}^2+\varepsilon_{13}^2}$	$\frac{L_c(\langle\sigma_{11}\rangle\langle\varepsilon_{11}\rangle+\sigma_{12}\varepsilon_{12}+\sigma_{13}\varepsilon_{13})}{\delta_{eq}^{ft}}$

Failure Mode	Equivalent Displacement (δeqi)	Equivalent Stress (σeqi)
FC	$L_c\langle-\varepsilon_{11}\rangle$	$\frac{L_c\langle-\sigma_{11}\rangle\langle-\varepsilon_{11}\rangle}{\delta_{eq}^{fc}}$
MT	$L_c\sqrt{\langle\varepsilon_{22}\rangle^2+\langle\varepsilon_{33}\rangle^2+\varepsilon_{12}^2+\varepsilon_{23}^2+\varepsilon_{13}^2}$	$\frac{L_c(\langle\sigma_{22}\rangle\langle\varepsilon_{22}\rangle+\langle\sigma_{33}\rangle\langle\varepsilon_{33}\rangle+\sigma_{12}\varepsilon_{12}+\sigma_{23}\varepsilon_{23}+\sigma_{13}\varepsilon_{13})}{\delta_{eq}^{mt}}$
MC	$L_c\sqrt{\langle-\varepsilon_{22}\rangle^2+\langle-\varepsilon_{33}\rangle^2+\varepsilon_{12}^2+\varepsilon_{23}^2+\varepsilon_{13}^2}$	$\frac{L_c(\langle-\sigma_{22}\rangle\langle-\varepsilon_{22}\rangle+\langle-\sigma_{33}\rangle\langle-\varepsilon_{33}\rangle+\sigma_{12}\varepsilon_{12}+\sigma_{23}\varepsilon_{23}+\sigma_{13}\varepsilon_{13})}{\delta_{eq}^{mc}}$

其中尖括号表示 Macaulay 括号：

$$\langle x\rangle=\begin{cases}x, & x>0\\ 0, & x\le 0\end{cases}$$

> 同样原文采用子程序也为vumat

这里再给出该文采用的退化矩阵与前文做对比，至于hashin准则可以自己去看

根据提供的文档，材料在损伤后发生刚度退化，其退化矩阵（包含损伤张量、柔度矩阵以及损伤刚度矩阵）的公式提取如下：

$$\mathbf{M}=\begin{bmatrix}1-d_f & 0 & 0 & 0 & 0 & 0\\ 0 & 1-d_m & 0 & 0 & 0 & 0\\ 0 & 0 & 1-d_m & 0 & 0 & 0\\ 0 & 0 & 0 & 1-d_s & 0 & 0\\ 0 & 0 & 0 & 0 & 1-d_s & 0\\ 0 & 0 & 0 & 0 & 0 & 1-d_s\end{bmatrix}$$

其中 d_f、d_m 和 d_s 分别表示纤维、基体和剪切模式的全局损伤变量，取值范围为0到1，计算公式为：

$$\begin{cases}d_f=1-(1-d_{ft})(1-d_{fc})\\ d_m=1-(1-d_{mt})(1-d_{mc})\\ d_s=1-(1-S_{mt}d_{mt})(1-S_{mc}d_{mc})\end{cases}$$

$$\mathbf{S}_d=\begin{bmatrix}\frac{1}{(1-d_f)E_{11}} & -\frac{\nu_{21}}{E_{22}} & -\frac{\nu_{31}}{E_{33}} & 0 & 0 & 0\\ -\frac{\nu_{12}}{E_{11}} & \frac{1}{(1-d_m)E_{22}} & -\frac{\nu_{32}}{E_{33}} & 0 & 0 & 0\\ -\frac{\nu_{13}}{E_{11}} & -\frac{\nu_{23}}{E_{22}} & \frac{1}{(1-d_m)E_{33}} & 0 & 0 & 0\\ 0 & 0 & 0 & \frac{1}{(1-d_s)G_{12}} & 0 & 0\\ 0 & 0 & 0 & 0 & \frac{1}{(1-d_s)G_{23}} & 0\\ 0 & 0 & 0 & 0 & 0 & \frac{1}{(1-d_s)G_{31}}\end{bmatrix}$$

损伤刚度矩阵 $\mathbf{C}_d$ 可通过对受损柔度矩阵 $\mathbf{S}_d$ 求逆获得（$\mathbf{C}_d=\mathbf{S}_d^{-1}$）。其矩阵形式为：

$$\mathbf{C}_d=\begin{bmatrix}dC_{11} & dC_{12} & dC_{13} & 0 & 0 & 0\\ dC_{12} & dC_{22} & dC_{23} & 0 & 0 & 0\\ dC_{13} & dC_{23} & dC_{33} & 0 & 0 & 0\\ 0 & 0 & 0 & dC_{44} & 0 & 0\\ 0 & 0 & 0 & 0 & dC_{55} & 0\\ 0 & 0 & 0 & 0 & 0 & dC_{66}\end{bmatrix}$$

矩阵中各项具体展开如下：

- $dC_{11}=(1-d_f)E_{11}[1-(1-d_m)^2\nu_{23}\nu_{32}]/\Delta$
- $dC_{22}=(1-d_m)E_{22}[1-(1-d_f)(1-d_m)\nu_{13}\nu_{31}]/\Delta$
- $dC_{33}=(1-d_m)E_{33}[1-(1-d_f)(1-d_m)\nu_{12}\nu_{21}]/\Delta$
- $dC_{12}=(1-d_f)(1-d_m)E_{11}[(1-d_m)\nu_{31}\nu_{23}+\nu_{21}]/\Delta$
- $dC_{13}=(1-d_f)(1-d_m)E_{11}[(1-d_m)\nu_{21}\nu_{32}+\nu_{31}]/\Delta$
- $dC_{23}=(1-d_m)^2E_{22}[(1-d_f)\nu_{12}\nu_{31}+\nu_{32}]/\Delta$

- $dC_{44} = (1 - d_s)G_{12}$
- $dC_{55} = (1 - d_s)G_{23}$
- $dC_{66} = (1 - d_s)G_{31}$

其中，分母参数 Δ 的表达式为：

$$\Delta = 1 - (1 - d_m)^2\nu_{32}\nu_{23} - (1 - d_f)(1 - d_m)\nu_{31}\nu_{13} - (1 - d_f)(1 - d_m)\nu_{21}\nu_{12} - 2(1 - d_f)(1 - d_m)^2\nu_{13}\nu_{21}\nu_{32}$$

> 该文的退化公式推导更严格，如果用该退化矩阵编写umat，看起来难收敛所以没用，参考它把前面Thuc剪切相中的（1-ds）去掉了

粘性正则化

来自Abaqus2024官方帮助手册，帮助收敛，hashin.for子程序使用了该粘性正则化

$$\dot{d}_v = \frac{1}{\eta}(d - d_v)$$

- d 是你根据之前讲的 Hashin 准则和断裂能 G^c 算出来的**真实损伤变量**（无粘性）。
- d_v 是系统实际用来降级刚度矩阵的**粘性损伤变量。**
- η 是**粘性系数 (Viscosity Coefficient)**，代表松弛时间。

图中的第二个公式：

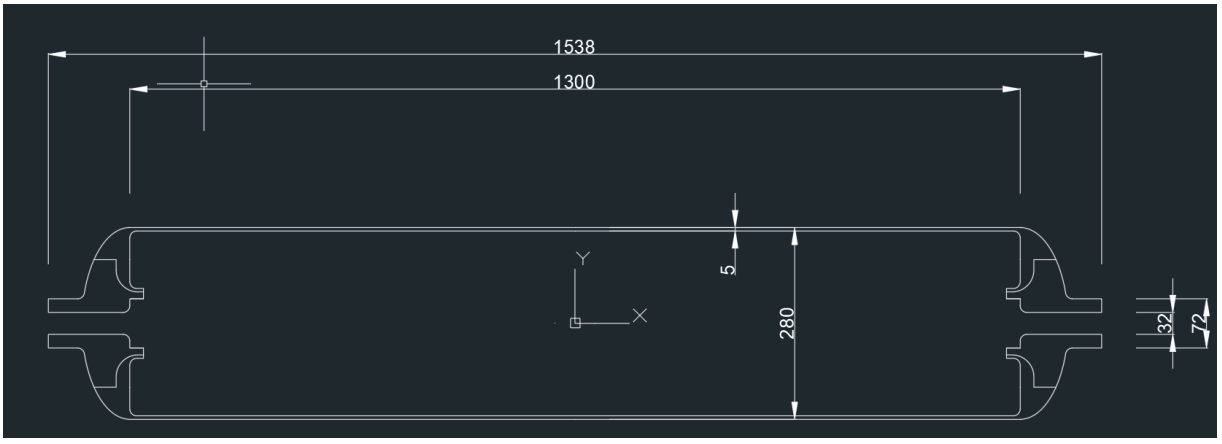
$$\sigma = \mathbf{C_d}\varepsilon$$

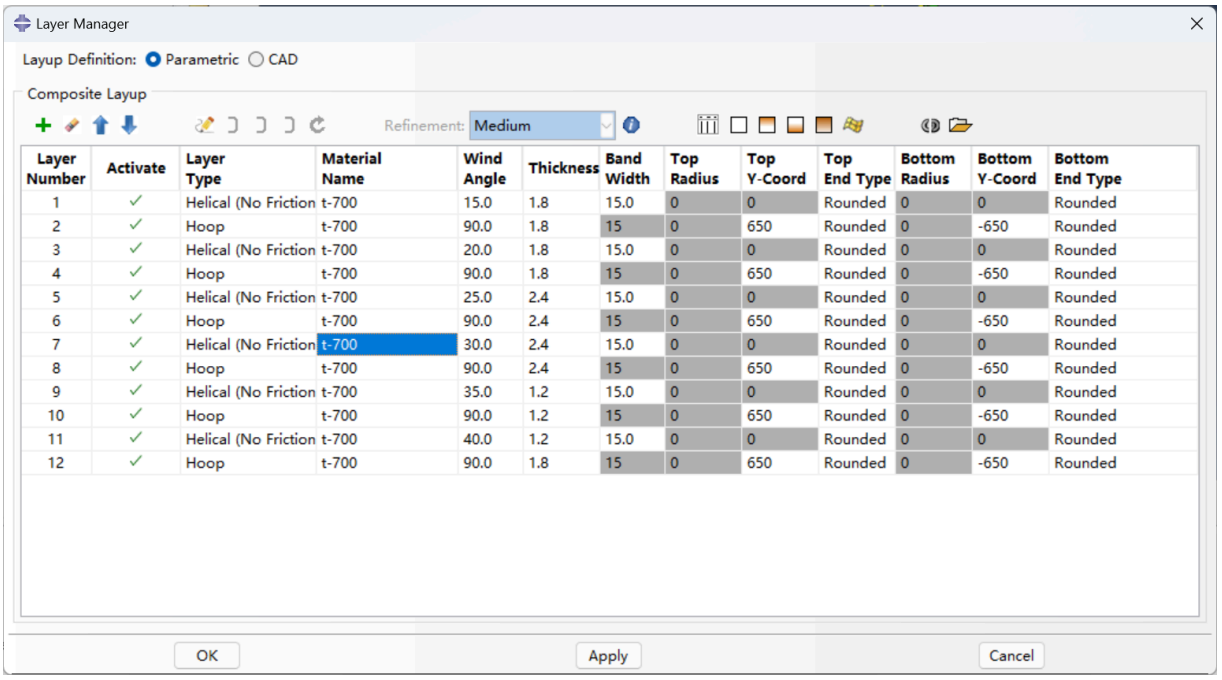
这里的 $\mathbf{C_d}$（受损弹性矩阵，就是我们之前长篇推导的那个大矩阵）里面的 d_f, d_m 等变量，**不再使用真实的 d，而是全部替换成了带有缓冲性质的 d_v。**

IV型瓶模型及所用参数

本文IV型瓶建模铺层及材料参数参考浙大硕士论文“IV 型高压储氢气瓶结构设计及优化研究”

为了验证子程序，个人在该模型及铺层上进行稍微改动





Layer Number	Activate	Layer Type	Material Name	Wind Angle	Thickness	Band Width	Top Radius	Top Y-Coord	Top End Type	Bottom Radius	Bottom Y-Coord	Bottom End Type
1	✓	Helical (No Friction	t-700	15.0	1.8	15.0	0	0	Rounded	0	0	Rounded
2	✓	Hoop	t-700	90.0	1.8	15	0	650	Rounded	0	-650	Rounded
3	✓	Helical (No Friction	t-700	20.0	1.8	15.0	0	0	Rounded	0	0	Rounded
4	✓	Hoop	t-700	90.0	1.8	15	0	650	Rounded	0	-650	Rounded
5	✓	Helical (No Friction	t-700	25.0	2.4	15.0	0	0	Rounded	0	0	Rounded
6	✓	Hoop	t-700	90.0	2.4	15	0	650	Rounded	0	-650	Rounded
7	✓	Helical (No Friction	t-700	30.0	2.4	15.0	0	0	Rounded	0	0	Rounded
8	✓	Hoop	t-700	90.0	2.4	15	0	650	Rounded	0	-650	Rounded
9	✓	Helical (No Friction	t-700	35.0	1.2	15.0	0	0	Rounded	0	0	Rounded
10	✓	Hoop	t-700	90.0	1.2	15	0	650	Rounded	0	-650	Rounded
11	✓	Helical (No Friction	t-700	40.0	1.2	15.0	0	0	Rounded	0	0	Rounded
12	✓	Hoop	t-700	90.0	1.8	15	0	650	Rounded	0	-650	Rounded

> 注意，这里模型只是测试案例，建模尤其封头及铺层角度与层数没有经过严格验算，只是顺手估计，并且所用材料等参数来源于不同文献，最终结果与实际无关

材料	密度 kg/m−3	抗拉强度/MPa	屈服强度/MPa	弹性模量/GPa	泊松比	导热系数 W/(m·K)	比热容 J/(g·K)
PA6	1130	55	47	1.88	0.4	0.27	1.5
6061-T6 铝合金	2700	340	306	70	0.32	218	0.9

参数	数值
弹性模量	$E_1 = 141\ \text{GPa},\ E_2 = 11.4\ \text{GPa},\ E_3 = 11.4\ \text{GPa}$
剪切模量	$G_{12} = 7.1\ \text{GPa},\ G_{13} = 7.1\ \text{GPa},\ G_{23} = 7.1\ \text{GPa}$
拉伸强度	$X_T = 2080\ \text{MPa},\ Y_T = 60\ \text{MPa}$
压缩强度	$X_C = 1250\ \text{MPa},\ Y_C = 290\ \text{MPa}$
剪切强度	$S_{12} = 110\ \text{MPa},\ S_{13} = 110\ \text{MPa},\ S_{23} = 110\ \text{MPa}$
泊松比	$\mu_{12} = 0.28,\ \mu_{13} = 0.28,\ \mu_{23} = 0.28$

Parameter	Value	Parameter	Value
E1/GPa	130	Xt/MPa	2080
E2/GPa	7.7	Xc/MPa	1250
E3/GPa	7.7	Yt/MPa	60
v12	0.3	Yc/MPa	140
v13	0.3	Sxy/MPa	110
v23	0.35	Gft/N/mm	133
G12/GPa	4.8	Gfc/N/mm	10

Parameter	Value	Parameter	Value
G13/GPa	4.8	Gmt/N/mm	0.5
G23/GPa	3.8	Gmc/N/mm	1.6
		Gs/N/mm	1.6

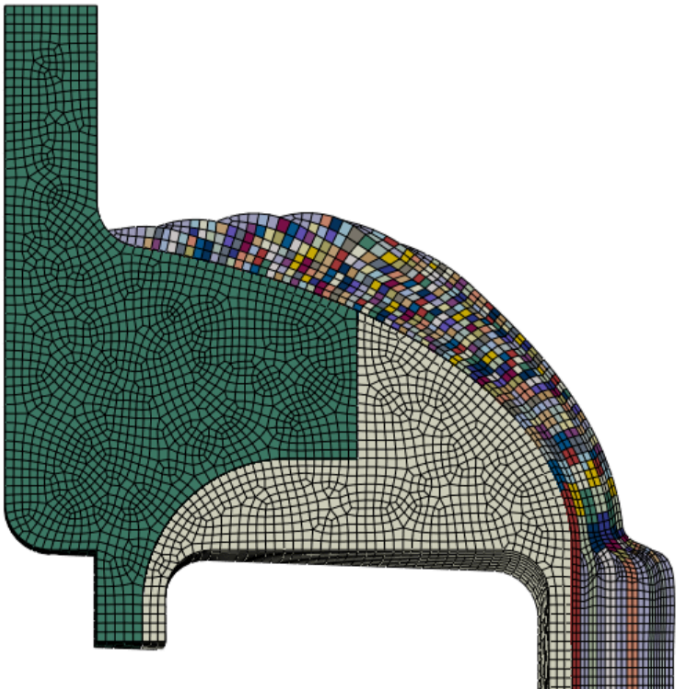
> **注意，第三个表来源于文献“Strength and Failure Analysis of Fiber-Wound Composite Gas Cylinder via Numerical Simulation”，这里只用它的断裂能参数，即gft-gmc**

Hashin.for 子程序运行及对应结果讨论

对应inp文件Job-hashinnewcdm.inp，所用模型见上

工况：直接使用的250MPa,观测最终爆炸失效压力

模型展示



部分sta文件

zhuningnew > hashincdm > Job-hashinnewcdm.sta

STEP	INC	ATT	SEVERE	EQUIL	TOTAL	TOTAL	STEP	INC OF	DOF	IF
1	85	1U	0	9	9	0.825	0.825	0.005000		
1	85	2	0	9	9	0.828	0.828	0.002500		
1	86	1	0	7	7	0.830	0.830	0.002500		
1	87	1	0	6	6	0.833	0.833	0.002500		
1	88	1	0	6	6	0.835	0.835	0.002500		
1	89	1	0	5	5	0.838	0.838	0.002500		
1	90	1	0	6	6	0.840	0.840	0.002500		
1	91	1U	0	6	6	0.840	0.840	0.002500		
1	91	2U	0	5	5	0.840	0.840	0.0006250		
1	91	3U	0	7	7	0.840	0.840	0.0001563		
1	91	4	0	5	5	0.840	0.840	3.906e-005		
1	92	1	0	6	6	0.840	0.840	3.906e-005		
1	93	1	0	3	3	0.840	0.840	3.906e-005		
1	94	1	0	3	3	0.840	0.840	3.906e-005		
1	95	1	0	4	4	0.840	0.840	5.859e-005		
1	96	1	0	3	3	0.840	0.840	8.789e-005		
1	97	1	0	4	4	0.840	0.840	0.0001318		
1	98	1U	0	4	4	0.840	0.840	0.0001978		
1	98	2U	0	6	6	0.840	0.840	4.944e-005		
1	98	3	0	4	4	0.840	0.840	1.236e-005		
1	99	1	0	4	4	0.840	0.840	1.854e-005		
1	100	1	0	3	3	0.840	0.840	2.781e-005		
1	101	1	0	2	2	0.841	0.841	4.171e-005		
1	102	1	0	3	3	0.841	0.841	6.257e-005		
1	103	1U	0	5	5	0.841	0.841	9.386e-005		
1	103	2	0	4	4	0.841	0.841	2.346e-005		
1	104	1	0	4	4	0.841	0.841	3.520e-005		
1	105	1U	0	4	4	0.841	0.841	5.279e-005		
1	105	2	0	4	4	0.841	0.841	1.320e-005		
1	106	1	0	4	4	0.841	0.841	1.980e-005		
1	107	1	0	3	3	0.841	0.841	2.970e-005		
1	108	1	0	3	3	0.841	0.841	4.454e-005		
1	109	1U	0	4	4	0.841	0.841	6.682e-005		
1	109	2	0	4	4	0.841	0.841	1.670e-005		
1	110	1	0	4	4	0.841	0.841	2.506e-005		
1	111	1	0	3	3	0.841	0.841	3.758e-005		
1	112	1U	0	4	4	0.841	0.841	5.638e-005		
1	112	2	0	3	3	0.841	0.841	1.409e-005		
1	113	1	0	4	4	0.841	0.841	2.114e-005		
1	114	1	0	2	2	0.841	0.841	3.171e-005		
1	115	1U	0	4	4	0.841	0.841	4.757e-005		
1	115	2	0	3	3	0.841	0.841	1.189e-005		
1	116	1	0	3	3	0.841	0.841	1.784e-005		
1	117	1	0	2	2	0.841	0.841	2.676e-005		
1	118	1	0	2	2	0.841	0.841	4.014e-005		
1	119	1U	0	8	8	0.841	0.841	6.020e-005		
1	131	2	0	3	3	0.841	0.841	7.628e-006		
1	132	1	0	3	3	0.841	0.841	1.144e-005		
1	133	1	0	2	2	0.841	0.841	1.716e-005		
1	134	1U	0	4	4	0.841	0.841	2.574e-005		
1	134	2	0	3	3	0.841	0.841	6.436e-006		
1	135	1	0	3	3	0.841	0.841	9.654e-006		
1	136	1	0	2	2	0.841	0.841	1.448e-005		
1	137	1	0	2	2	0.841	0.841	2.172e-005		
1	138	1U	0	6	6	0.841	0.841	3.258e-005		
1	138	2	0	3	3	0.841	0.841	8.146e-006		
1	139	1	0	3	3	0.841	0.841	1.222e-005		
1	140	1	0	2	2	0.841	0.841	1.833e-005		
1	141	1U	0	5	5	0.841	0.841	2.749e-005		
1	141	2	0	3	3	0.841	0.841	6.873e-006		
1	142	1	0	3	3	0.841	0.841	1.031e-005		
1	143	1	0	2	2	0.841	0.841	1.546e-005		
1	144	1U	0	5	5	0.841	0.841	2.320e-005		
1	144	2	0	3	3	0.841	0.841	5.799e-006		
1	145	1	0	3	3	0.841	0.841	8.699e-006		
1	146	1	0	2	2	0.841	0.841	1.305e-005		
1	147	1U	0	4	4	0.841	0.841	1.957e-005		
1	147	2	0	2	2	0.841	0.841	4.893e-006		
1	148	1	0	3	3	0.841	0.841	7.340e-006		
1	149	1	0	2	2	0.841	0.841	1.101e-005		
1	150	1	0	1	1	0.841	0.841	1.651e-005		
1	151	1U	0	5	5	0.841	0.841	2.477e-005		
1	151	2	0	3	3	0.841	0.841	6.193e-006		
1	152	1	0	3	3	0.841	0.841	9.289e-006		
1	153	1	0	16	16	0.841	0.841	1.393e-005		
1	154	1	0	2	2	0.841	0.841	1.045e-005		

THE ANALYSIS HAS NOT BEEN COMPLETED

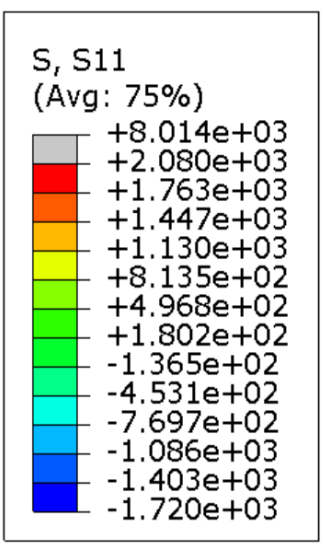
> 大约在0.84处出现震荡，估计该模型最终爆压失效为210MPa(别当真，明显铺层铺厚了)

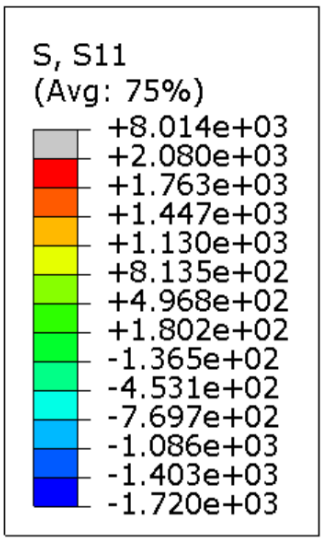
SDV对应含义

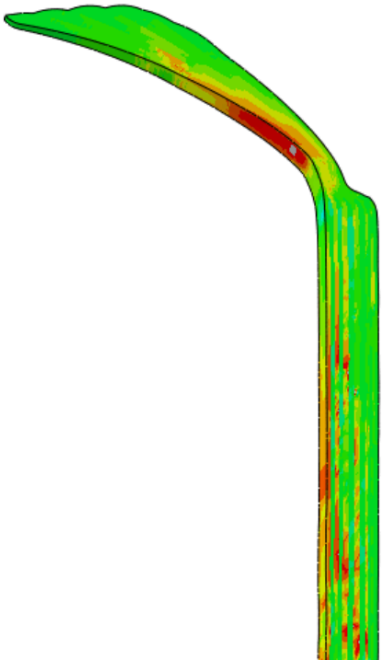
- SDV1 = dft，纤维拉伸损伤。
- SDV2 = dfc，纤维压缩损伤。
- SDV3 = dmt，基体拉伸/剪切损伤。
- SDV4 = dmc，基体压缩/剪切损伤。
- SDV5~SDV8 = 四种模式对应的历史变量 kappa，代码注释写的是等效应变，但实际是按 CELENT*应变 构造的等效位移型历史量。[
- SDV9~SDV12 = 失效起始标志，某模式一旦达到 Hashin 判据 FI>=1，对应标志就置 1。
- SDV13~SDV16 = 损伤起始时的等效位移 DELTA0。
- SDV17~SDV20 = 损伤起始时的等效应力 SIGMA0。
- SDV21~SDV24 = 黏性正则化后的损伤变量 dftv/dfcv/dmtv/dmcv，这四个才是实际参与刚度退化的量。

实际云图

s11

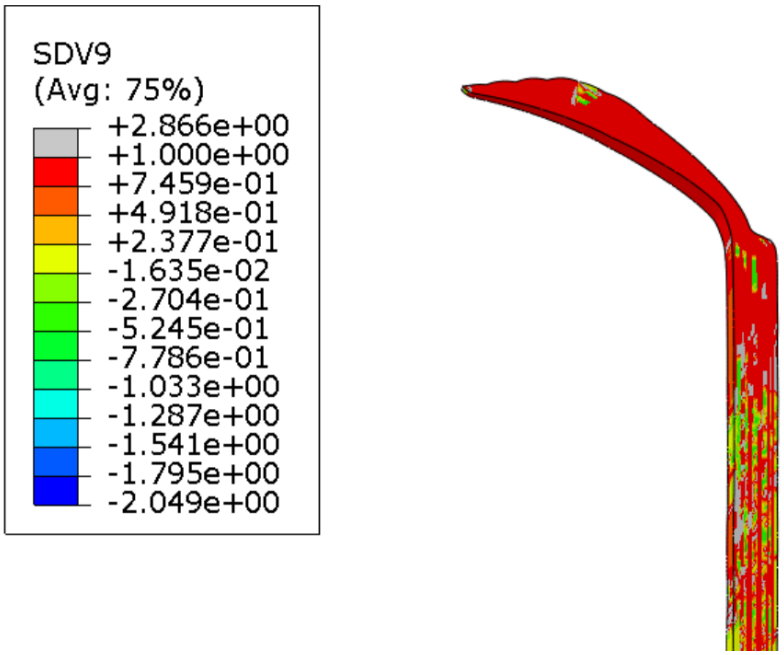


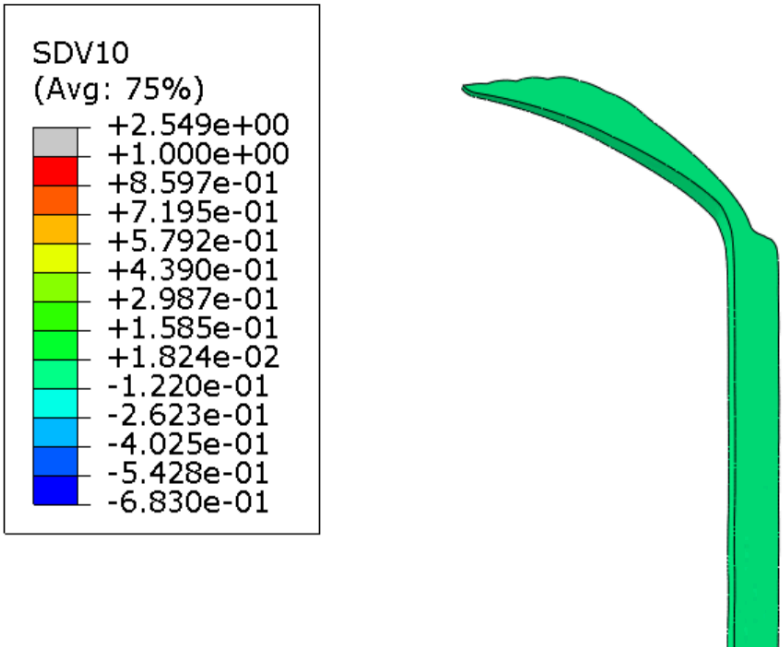


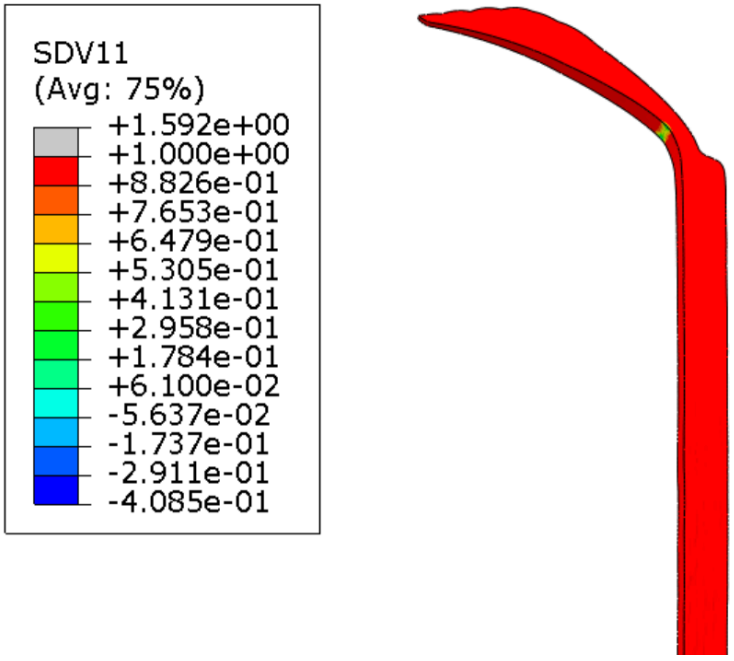


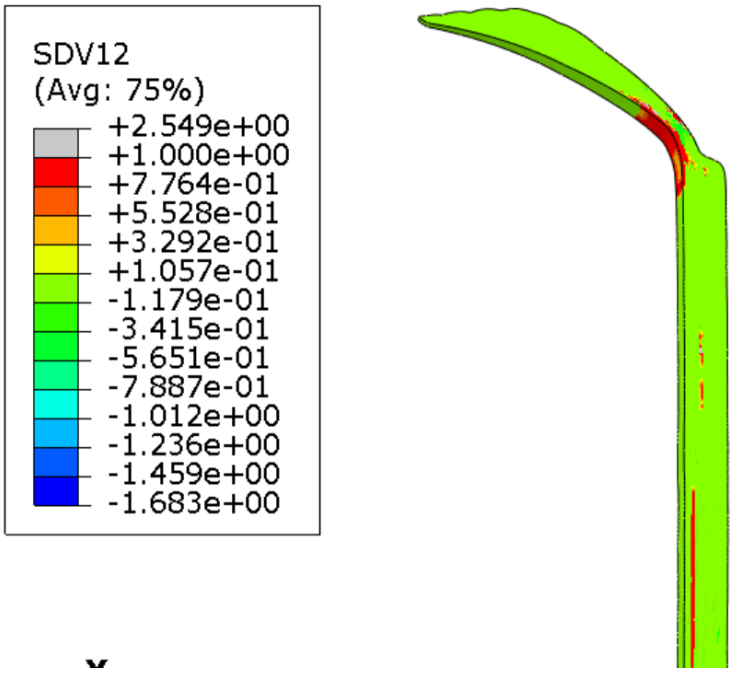
SDV 9-12

> 最终失效结果，要看哪个部分刚起裂，往前面调，这里做演示


SDV9
(Avg: 75%)
+2.866e+00
+1.000e+00
+7.459e-01
+4.918e-01
+2.377e-01
-1.635e-02
-2.704e-01
-5.245e-01
-7.786e-01
-1.033e+00
-1.287e+00
-1.541e+00
-1.795e+00
-2.049e+00

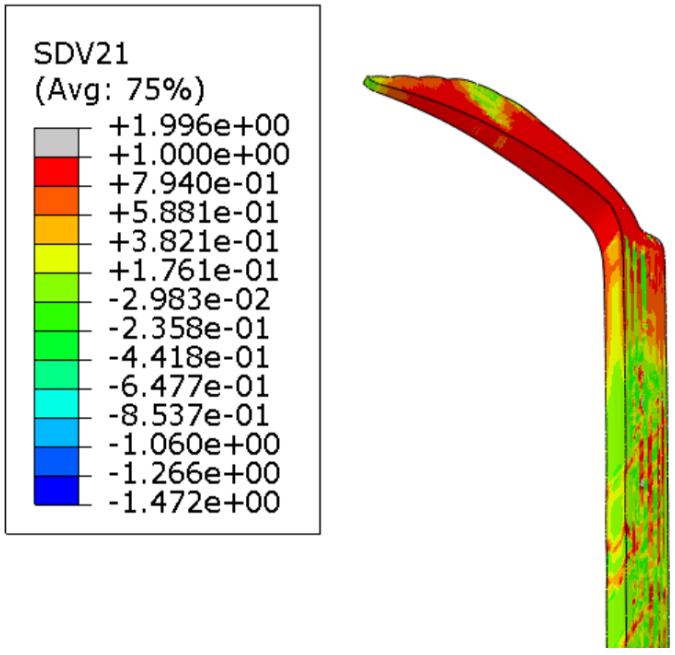

SDV10
(Avg: 75%)
+2.549e+00
+1.000e+00
+8.597e-01
+7.195e-01
+5.792e-01
+4.390e-01
+2.987e-01
+1.585e-01
+1.824e-02
-1.220e-01
-2.623e-01
-4.025e-01
-5.428e-01
-6.830e-01

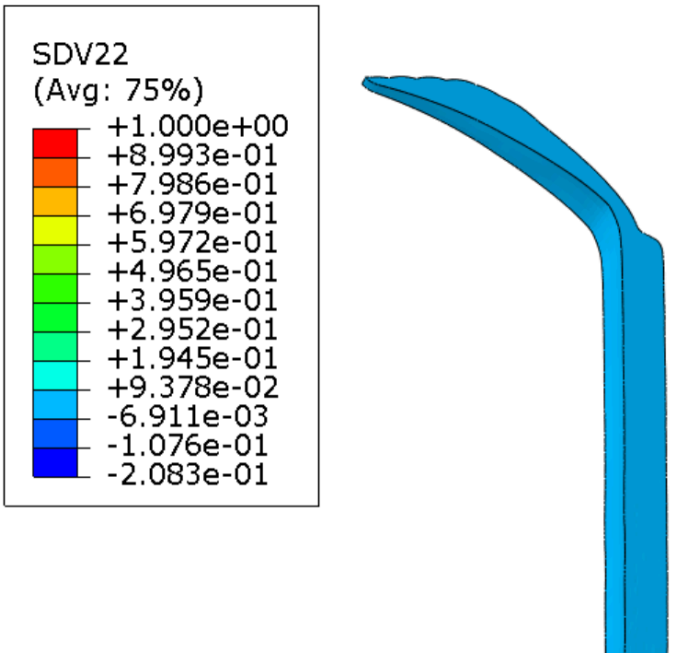

SDV11
(Avg: 75%)
+1.592e+00
+1.000e+00
+8.826e-01
+7.653e-01
+6.479e-01
+5.305e-01
+4.131e-01
+2.958e-01
+1.784e-01
+6.100e-02
-5.637e-02
-1.737e-01
-2.911e-01
-4.085e-01

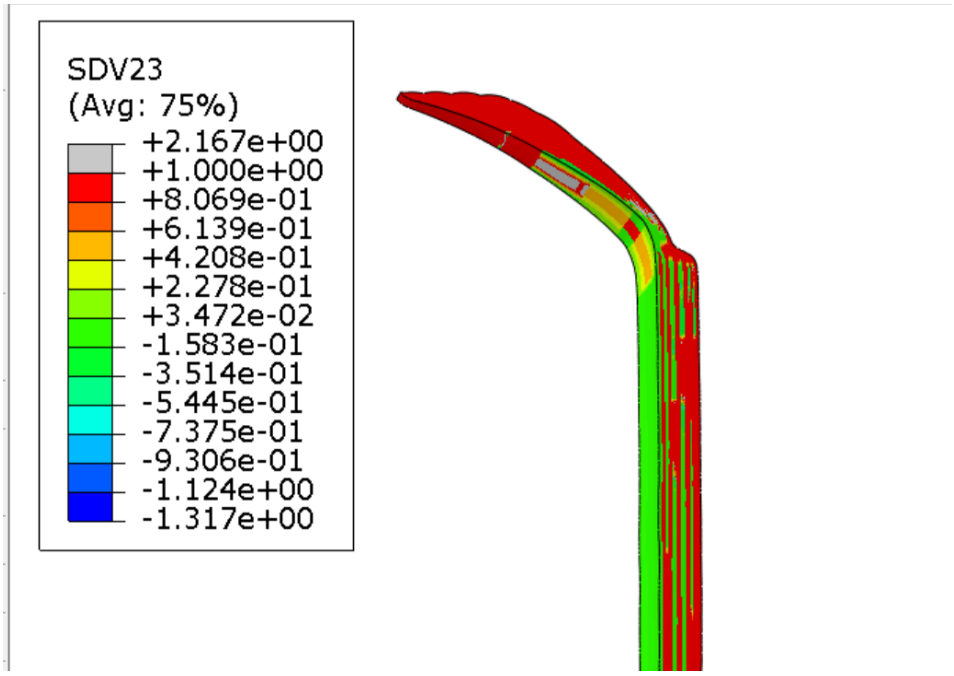


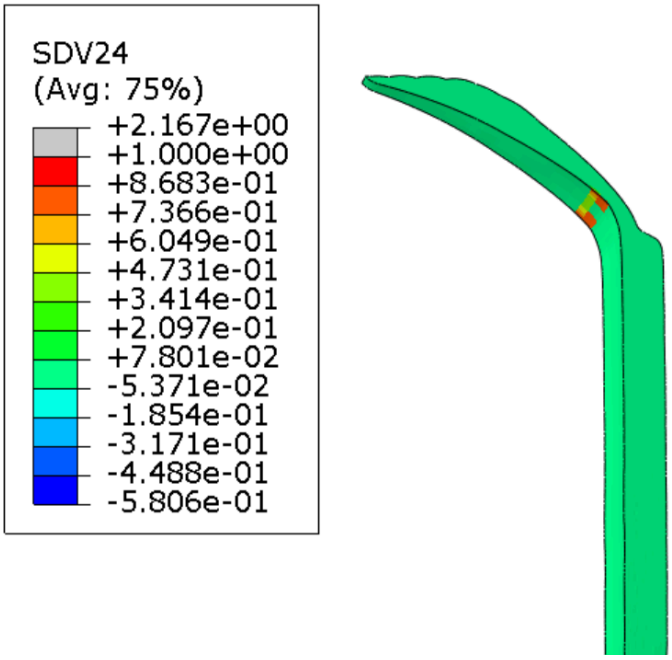
SDV 21-24

> 注意该子程序调整了纤维树脂失效阈值
>
> - 纤维损伤 dft/dfc 上限 0.95
> - 树脂/基体损伤 dmt/dmc 上限 0.85
>
> 真正参与刚度退化的是黏性后的变量和组合变量，不是原始 df：
>
> - 纤维综合损伤 dfv = 1-(1-dftv)(1-dfcv)，上限 0.95
> - 基体综合损伤 dmv = 1-(1-dmtv)(1-dmcv)，上限 0.85
> **如果觉得不合适也是自己调，这里给树脂更大的冗余是因为树脂最先失效，防止其影响收敛**




SDV22
(Avg: 75%)
+1.000e+00
+8.993e-01
+7.986e-01
+6.979e-01
+5.972e-01
+4.965e-01
+3.959e-01
+2.952e-01
+1.945e-01
+9.378e-02
-6.911e-03
-1.076e-01
-2.083e-01


SDV23
(Avg: 75%)
+2.167e+00
+1.000e+00
+8.069e-01
+6.139e-01
+4.208e-01
+2.278e-01
+3.472e-02
-1.583e-01
-3.514e-01
-5.445e-01
-7.375e-01
-9.306e-01
-1.124e+00
-1.317e+00


SDV24
(Avg: 75%)
+2.167e+00
+1.000e+00
+8.683e-01
+7.366e-01
+6.049e-01
+4.731e-01
+3.414e-01
+2.097e-01
+7.801e-02
-5.371e-02
-1.854e-01
-3.171e-01
-4.488e-01
-5.806e-01

关于hashin_constant.for及其结果讨论

> 这是直接刚度折减，即直接给出确定的df等损伤值
>
> 另外所用公式为文献1的原始公式

对应折减系数来源文献“Progressive damage analysis for multiscale modelling of composite pressure vessels based on Puck failure criterion”

Contents lists available at ScienceDirect

Composite Structures

journal homepage: www.elsevier.com/locate/compstruct

Progressive damage analysis for multiscale modelling of composite pressure vessels based on Puck failure criterion

Song Lin [a], Liuqing Yang [a], Hui Xu [d], Xiaolong Jia [b,c,*], Xiaoping Yang [b,c], Lei Zu [d,*]

Loading-type failure category	Mesoscopic failure mode	Micro-stiffness reduction coefficient
Tensile failure	Fibre tensile failure	$E_{f11d} = 0.07E_{f11}$ $E_{f22d} = E_{f33d} = 0.07E_{f22}$ $\mu_{f12d} = 0.07\mu_{f12}$ $\mu_{f23d} = 0.07\mu_{f23}$ $G_{f12d} = G_{f13d} = 0.07G_{f12}$ $G_{f23d} = 0.07G_{f23}$
	Matrix tensile failure	$E_{md} = 0.2E_m$ $\mu_{md} = 0.2\mu_m$ $G_{md} = 0.2G_m$
	Fibre–matrix tensile failure	Ditto
Compression failure	Fibre compression failure	$E_{f11d} = 0.14E_{f11}$ $E_{f22d} = E_{f33d} = 0.14E_{f22}$ $\mu_{f12d} = 0.14\mu_{f12}$ $\mu_{f23d} = 0.14\mu_{f23}$ $G_{f12d} = G_{f13d} = 0.14G_{f12}$ $G_{f23d} = 0.14G_{f23}$
	Matrix compression failure	$E_{md} = 0.4E_m$ $\mu_{md} = 0.4\mu_m$ $G_{md} = 0.4G_m$
	Fibre–matrix compression failure	Ditto

STA文件

zhuningnew > hashinconstant > Job-hashinnewconstant.sta

STEP	INC	ATT	SEVERE	EQUIL	TOTAL	TOTAL	STEP	INC OF	D(
1	77	1	0	2	2	0.770	0.770	0.01000	
1	78	1	0	2	2	0.780	0.780	0.01000	
1	79	1	0	2	2	0.790	0.790	0.01000	
1	80	1	0	2	2	0.800	0.800	0.01000	
1	81	1	0	2	2	0.810	0.810	0.01000	
1	82	1	0	2	2	0.820	0.820	0.01000	
1	83	1	0	2	2	0.830	0.830	0.01000	
1	84	1	0	2	2	0.840	0.840	0.01000	
1	85	1	0	2	2	0.850	0.850	0.01000	
1	86	1	0	2	2	0.860	0.860	0.01000	
1	87	1	0	2	2	0.870	0.870	0.01000	
1	88	1	0	2	2	0.880	0.880	0.01000	
1	89	1	0	2	2	0.890	0.890	0.01000	
1	90	1	0	2	2	0.900	0.900	0.01000	
1	91	1	0	2	2	0.910	0.910	0.01000	
1	92	1	0	2	2	0.920	0.920	0.01000	
1	93	1	0	2	2	0.930	0.930	0.01000	
1	94	1	0	2	2	0.940	0.940	0.01000	
1	95	1	0	2	2	0.950	0.950	0.01000	
1	96	1	0	2	2	0.960	0.960	0.01000	
1	97	1	0	2	2	0.970	0.970	0.01000	
1	98	1	0	2	2	0.980	0.980	0.01000	
1	99	1	0	2	2	0.990	0.990	0.01000	
1	100	1	0	2	2	1.00	1.00	0.01000	

THE ANALYSIS HAS COMPLETED SUCCESSFULLY

SDV

- SDV1 = FI(1)，纤维拉伸失效指数
- SDV2 = FI(2)，纤维压缩失效指数
- SDV3 = FI(3)，基体拉伸失效指数
- SDV4 = FI(4)，基体压缩失效指数
- SDV5 = df，综合纤维损伤
- SDV6 = dm，综合基体损伤
- SDV7 = DMG(1)，纤维拉伸是否触发，0/1
- SDV8 = DMG(2)，纤维压缩是否触发，0/1
- SDV9 = DMG(3)，基体拉伸是否触发，0/1
- SDV10 = DMG(4)，基体压缩是否触发，0/1

这套材料的固定退化参数来自 PROPS(17:22)：

- dft=0.07
- dfc=0.14
- dmt=0.2
- dmc=0.4
- smt=0.9
- smc=0.5

先看 SDV1~SDV4
这些是失效指数，>=1 就说明该模式达到 Hashin 触发条件。

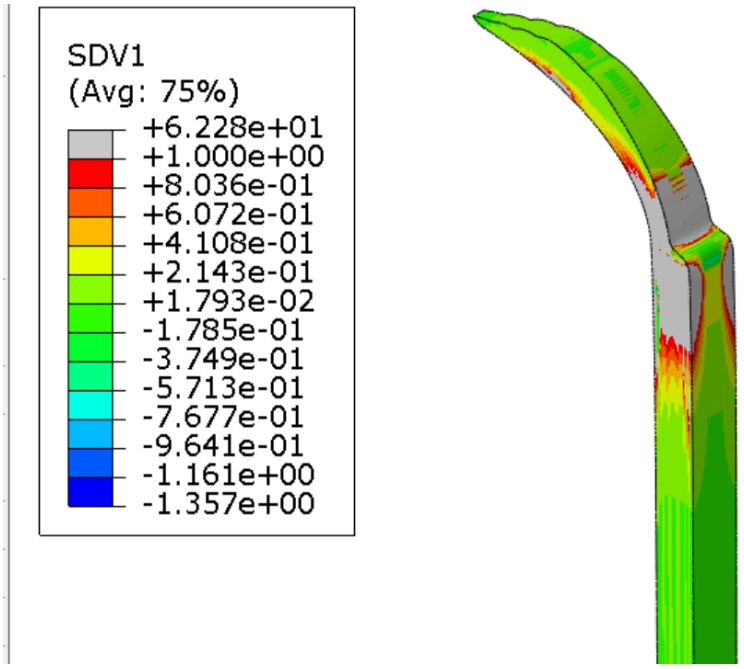
再看 SDV7~SDV10
这些是最直观的“哪种模式已经发生”，一旦触发就变成 1。

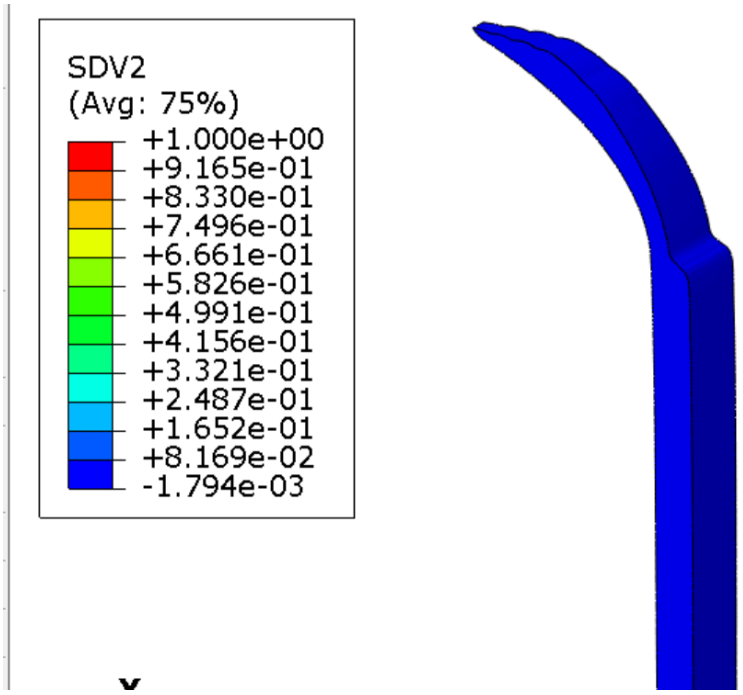
最后看 SDV5、SDV6
这两个是综合后的纤维/基体损伤，真正对应刚度退化强弱。

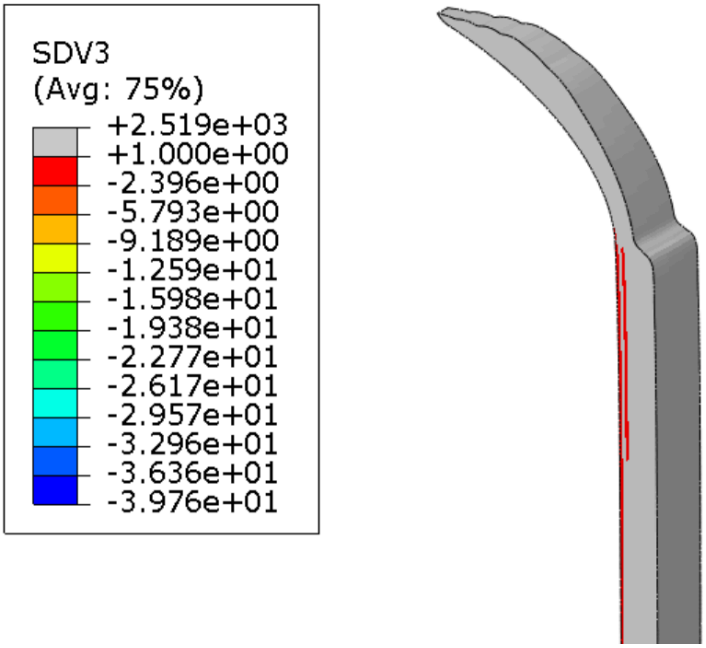
所以比如：

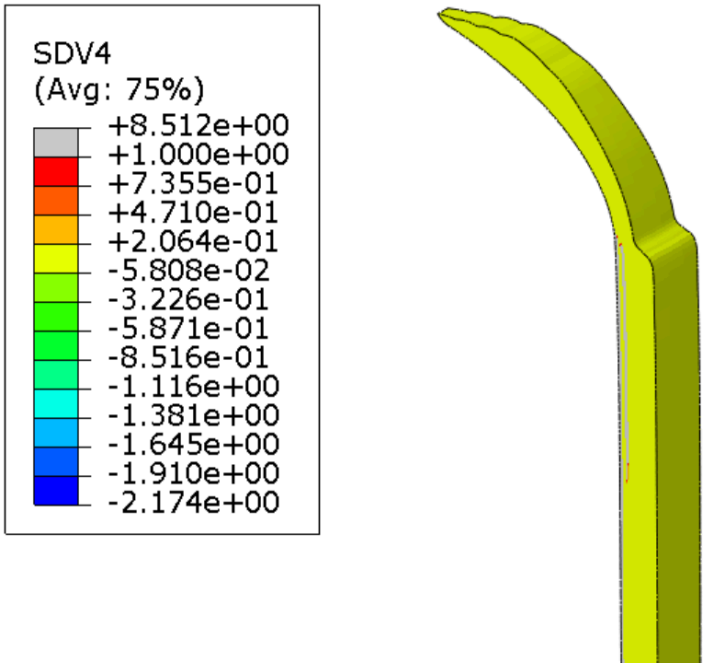
- **如果只发生纤维拉伸失效，SDV7=1，那 SDV5=df=0.07**
- **如果只发生纤维压缩失效，SDV8=1，那 SDV5=df=0.14**
- **如果只发生基体拉伸失效，SDV9=1，那 SDV6=dm=0.2**
- **如果只发生基体压缩失效，SDV10=1，那 SDV6=dm=0.4**

SDV1-4

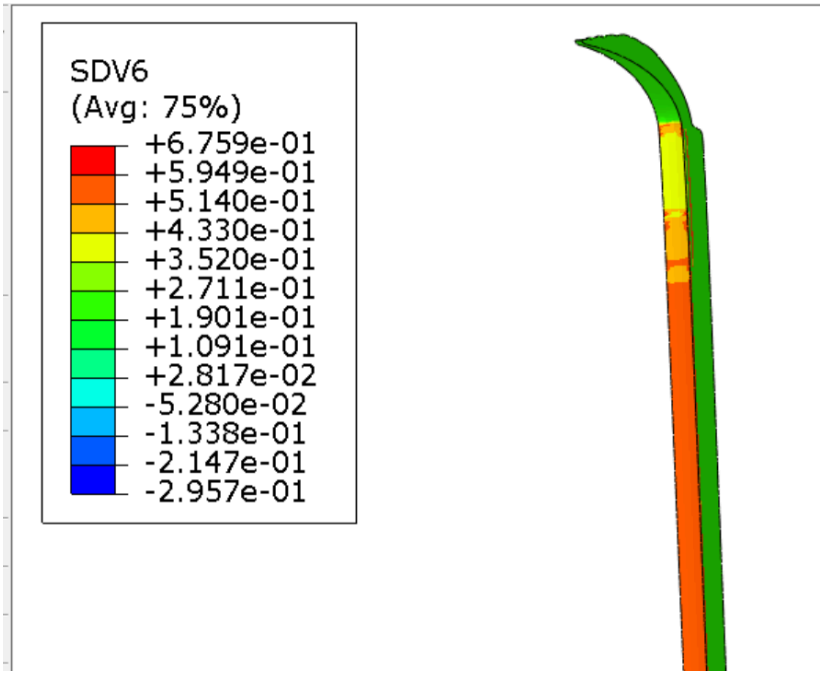





SDV3
(Avg: 75%)
+2.519e+03
+1.000e+00
-2.396e+00
-5.793e+00
-9.189e+00
-1.259e+01
-1.598e+01
-1.938e+01
-2.277e+01
-2.617e+01
-2.957e+01
-3.296e+01
-3.636e+01
-3.976e+01


SDV4
(Avg: 75%)
+8.512e+00
+1.000e+00
+7.355e-01
+4.710e-01
+2.064e-01
-5.808e-02
-3.226e-01
-5.871e-01
-8.516e-01
-1.116e+00
-1.381e+00
-1.645e+00
-1.910e+00
-2.174e+00

sdv6


SDV6
(Avg: 75%)
+6.759e-01
+5.949e-01
+5.140e-01
+4.330e-01
+3.520e-01
+2.711e-01
+1.901e-01
+1.091e-01
+2.817e-02
-5.280e-02
-1.338e-01
-2.147e-01
-2.957e-01